

Autonomous Trucking

Driver Out, Production Ramp Ahead. Physical AI Update After Rides & Meetings with Aurora and Kodiak

We recently attended the Aurora investor day (AUR, Not Covered) and also met with the management team at Kodiak AI (KDK, Not Covered), and after this week we are up to 11 rides in autonomous trucking systems but Aurora's was the first without any safety driver or engineer in the cab. The technology has undoubtedly come a long way since our first test rides on closed courses in 2019 (details [here](#)) and our autonomous week in 2022 (details [here](#)), to the point where the capability and safety of the technology is no longer routinely questioned. The self-driving truck industry has matured to the point where the key concerns are production timelines, commercial adoption and unit economics as Aurora projects exponential growth through 2030 while Volvo Autonomous Solutions (powered by the Aurora Driver) was the first legacy manufacturer to publicly announce targets for fleet size, revenue and unit economics in June 2026. We do not expect the first autonomous truck deliveries from Aurora or Volvo to materially disrupt the operations or revenues of conventional truckers or IMCs, but we will closely monitor the industry's ability to meet production targets in 2027 and maintain the high utilization rates needed to sustainably lower unit costs. Physical AI has emerged as an important investment theme and we will host six autonomous freight companies at our recently announced Future of Freight event as part of the J.P. Morgan U.S. Opportunities Forum on Nov 12-13 in Miami. Participants include Kodiak AI, PlusAI, TORC Robotics, Parallel Systems, Einride, and Gatik, contact your sales representative [here](#) to sign up. See below for highlights from our meetings and rides with Aurora and Kodiak.

- **Significant scaling of Aurora Driver fleet targeted by 2030.** The Aurora investor day framed the forecasted ramp up of autonomous trucks as 20+ years in the making, drawing parallels to the long investment cycles followed by rapid adoption in passenger automobiles, the internet and containerized shipping. Aurora plans to own and operate a fleet of < 500 trucks under the Transportation-as-a-Service to control operations and secure commitments from customers with retrofitted Gen 2 trucks. This path effectively bootstraps the Aurora Driver into the market with the goal of transitioning customers to owning and operating trucks from multiple OEMs that are line-fit with auto grade, Gen 3 hardware from AUMOVIO. Aurora is targeting 200 trucks exiting 2026 which grows to over 1,000 in 2027, roughly 7,500 in 2028, ~18,000 in 2029, and more than 30,000 by 2030 for a CAGR of 250%+.
- **Autonomous a potential long term negative for traditional freight.** At this point, we believe it is still too early to expect a material impact on freight rates from autonomous capacity entering the market given the industry's limited disclosure on adoption pace, lane breadth, and real-world utilization outside of early corridors. The first impact would likely be most noticeable in certain local corridors, although the risk could increase if large-scale shippers generate utilization gains and create a durable cost advantage that competitors are forced to match. This could also be a headwind for intermodal if over-the-road becomes faster, cheaper, and more reliable on key lanes, which could also impact LTL line-haul and dedicated networks. For comparison, with 30k trucks at an average of 250k miles per truck per year, Aurora expects to capture ~1.9% of the 400B mile long haul trucking SAM by 2030.
- **Success takes a supply chain, including insurance.** The chief underwriter of

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Apollo, Chris Moore, painted a supportive picture for insurance costs over the long run as miles accumulate in the system and accident modeling improves. Claim severity remains a key unknown with limited precedent in U.S. courtrooms, which is why autonomous trucking premiums are above comparable human fleets. Moore expects premiums to meaningfully decline as claims data builds, citing roughly 15–25% annual decreases, and emphasized that underwriting also hinges on safety culture, transparency, and readiness to defend against litigation. Insurance costs could remain elevated during the initial adoption phase because industry underwriting capacity has not yet expanded to focus on autonomous truck networks.

- **Customer adoption is a critical driver for the industry...** The technology behind self-driving trucks is increasingly viewed as a safer option for an industry responsible for 5k+ fatal crashes per year, per the latest annual data from the Insurance Institute for Highway Safety. Aurora is already hauling commercial freight for customers, reaching 500k fully driverless miles, while Kodiak plans to launch a driverless service between Dallas and Houston by the end of 2026. However, the self-driving systems will need to plug into existing carrier networks and supply chains as a starting point toward broader commercial adoption and scale. The ability to reach mutually acceptable pricing and unit economics with customers could also impact the pace of adoption considering Werner highlighted at the Aurora event that there appears to be a “gap” when it comes to finalizing contract terms.
- **... as well as mass production down the road.** Producing autonomous trucks at scale is a critical catalyst for the adoption flywheel, considering adoption can only go so far if it is fully dependent on upfitting self-driving kits. Successful production at scale also depends on upstream component availability and validation, including compute supply from Nvidia and the broader ecosystem of safety-critical redundancy components such as braking and steering systems from Tier 1 suppliers. Overall, scaling from tens of trucks to tens of thousands of trucks will require consistent and frequent communication across the entire supply chain, beginning well in advance of any material ramp up in production.
- **Kodiak AI facility visit and test ride.** We visited Kodiak AI’s operations center and terminal in the Dallas/Fort-Worth area and went on an autonomous ride on I-20 and I-45 with a safety driver. On Friday, Kodiak also announced that it plans to launch unsupervised, long-haul driverless trucking service on the Dallas–Houston freight lane by the end of 2026 after starting the launch program in August. The routes are run with safety drivers as part of final testing and validation for its initial long-haul safety case, while Kodiak is also conducting closed-course test track trials to stress-test rare or risky scenarios. The company also shared their Assisted Autonomous Driving technology where the vehicle can request assistance with unlearned obstacles from a CDL driver who can send instructions for the truck to execute on its own at speeds below 25mph. This technology seems like a critical part of Kodiak’s commercial strategy, but we have not yet ascertained how widely a similar type of technology is utilized across the industry.
- **Common questions and investor perspectives.** Autonomous trucking has been a more frequent discussion topic with investors over the last three months. Familiarity with self-driving passenger cars appears to be driving these conversations as the technology is viewed as increasingly capable and not limited by regulations at this point. With this backdrop, it is a positive sign for the self-driving truck industry that investor discussions are increasingly focused on the all-in economics of an autonomy-enabled tractor, specifically what it will cost a fleet to buy an autonomous-ready truck from an OEM versus a traditional truck and how that incremental capex (or embedded per-mile hardware cost) is shared across the ecosystem. The cost should come down as production scales up, and neither side is effectively subsidizing early development costs, but based on Werner’s comments noted above, there is still a pricing gap to close, which could impact the pace of adoption and commercial deployment. In another sign of the industry’s maturity, we also heard questions around the practical implications of higher utilization

rates on maintenance and warranties, which would be a good problem for a fleet to have but a potential challenge nevertheless. Investors are unsurprisingly still focused on cash burn and potential dilution from equity raises, which puts an increased importance on a successful and efficient production ramp. We believe the market would also like to see new customer announcements with firm delivery commitments to support the widely anticipated commercial inflection in 2027 and beyond.

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